

ARCH 594/508: AI Research Seminar/Studio—AI in Architecture
Spring 2026

Exploratory Data Analysis

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TODAY'S AGENDA

1. Exploratory Data Analysis (EDA)
2. Lab Activity: (EDA)

REVIEW

TYPES OF DATA

- **Categorical (Qualitative) Data:**

- Observable but not measurable attributes: colors, textures, smells, tastes.

- **Numerical (Quantitative) Data:**

- Measurable attributes: speed, age, temperature, height.

Discrete Data:

- Takes specific values (e.g., 1, 2, 3...).

Continuous Data:

- Can take any value within a particular range (e.g., [3.1, 8]).



REVIEW

TEMPORAL AND SPATIAL DIMENSIONS OF DATA

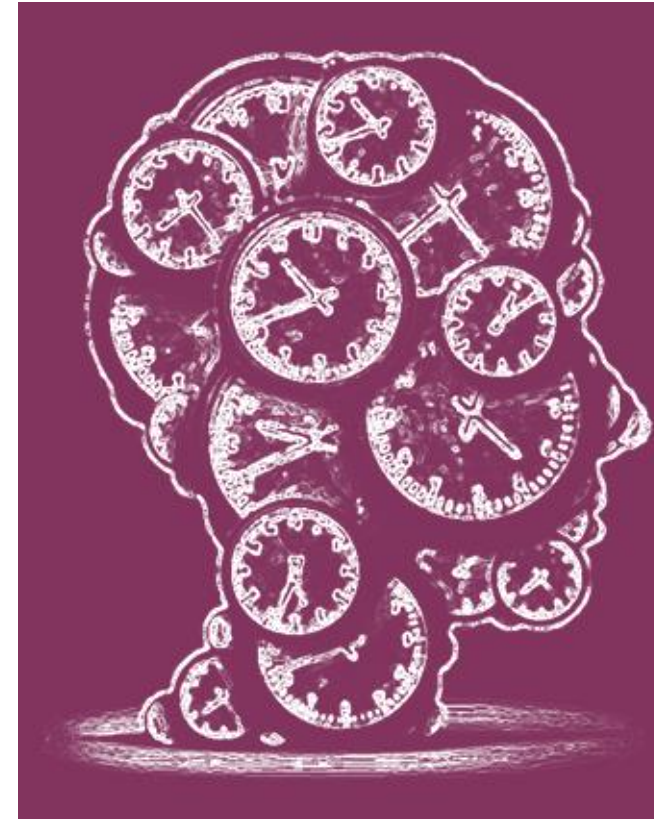
Temporal/ Time

■ **Static Data:**

- No variation over time.
- Example: Annual mean temperature in Seattle in 2023.

■ **Dynamic Data** (Time Series or Temporal):

- Varied data over time.
- Example: Hourly relative humidity in Seattle during January 2024.



REVIEW

TEMPORAL AND SPATIAL DIMENSIONS OF DATA

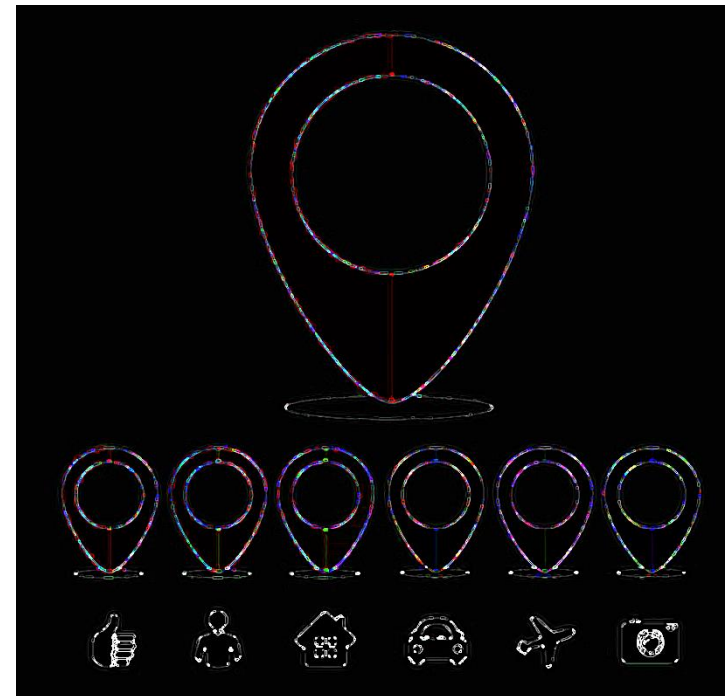
Spatial/Location

■ Non-Geospatial Data:

- May or may not include geo-referenced information.
- Cannot be mapped due to formatting or lack of geographic coordination information.

■ Geospatial Data:

- Includes geo-referenced information, mappable data.
- **Types:** Point-Location, Zonal Boundaries, Hypothetical Boundaries, Spatiotemporal.



REVIEW

DATA IN BUILT ENVIRONMENT



Structured Databases: Stored in tables for easy querying and analysis; public data



Unstructured Data (Text): Extracting insights from social media, emails, news



Big Data: Generated from various sources such as Public Wi-fi, Smartphone Apps, etc.)



Sensor Readings: IoT sensor data (e.g., temperature, mobility) for monitoring environments, cities, industries, and healthcare.



Simulated Data: Physics-based engineering simulations.



Other Generated Data: Derived from various design layouts and scenarios.

REVIEW

PYTHON DATA TYPES (CORE CONCEPTS)

- **Numeric:**

int (integers / $x = 10$), float (decimals / $y = 3.14$), complex (complex numbers / $z = 2 + 3j$)

- **Sequence (ordered):**

List ($lst = [1, 2, 3]$) , tuple ($t = (1, 2, 3)$) , range ($r = range(5)$)

- **Text:**

str — strings for textual data ($name = "AI Studio"$)

- **Boolean:**

bool — True / False for logic and conditions ($is_valid = True$)

KEY TERMINOLOGY

Key terminology

| Term | Meaning |

|---|---|

| **Variable** | A named container that stores a value (e.g., `x = 5` stores the number 5 in `x`) |

| **DataFrame** | A table of data with rows and columns — the pandas equivalent of a spreadsheet |

| **Column** | A single vertical field in a table (e.g., "Age" or "Name") |

| **Row** | A single horizontal entry in a table (e.g., one building or one animal) |

| **Function** | A reusable instruction, such as `print()`, which displays text on screen |

| **Index** | The position number of an item — Python counts starting from **0**, not 1 |

| **Filter** | Keeping only the rows that meet a condition (e.g., "age > 30") |

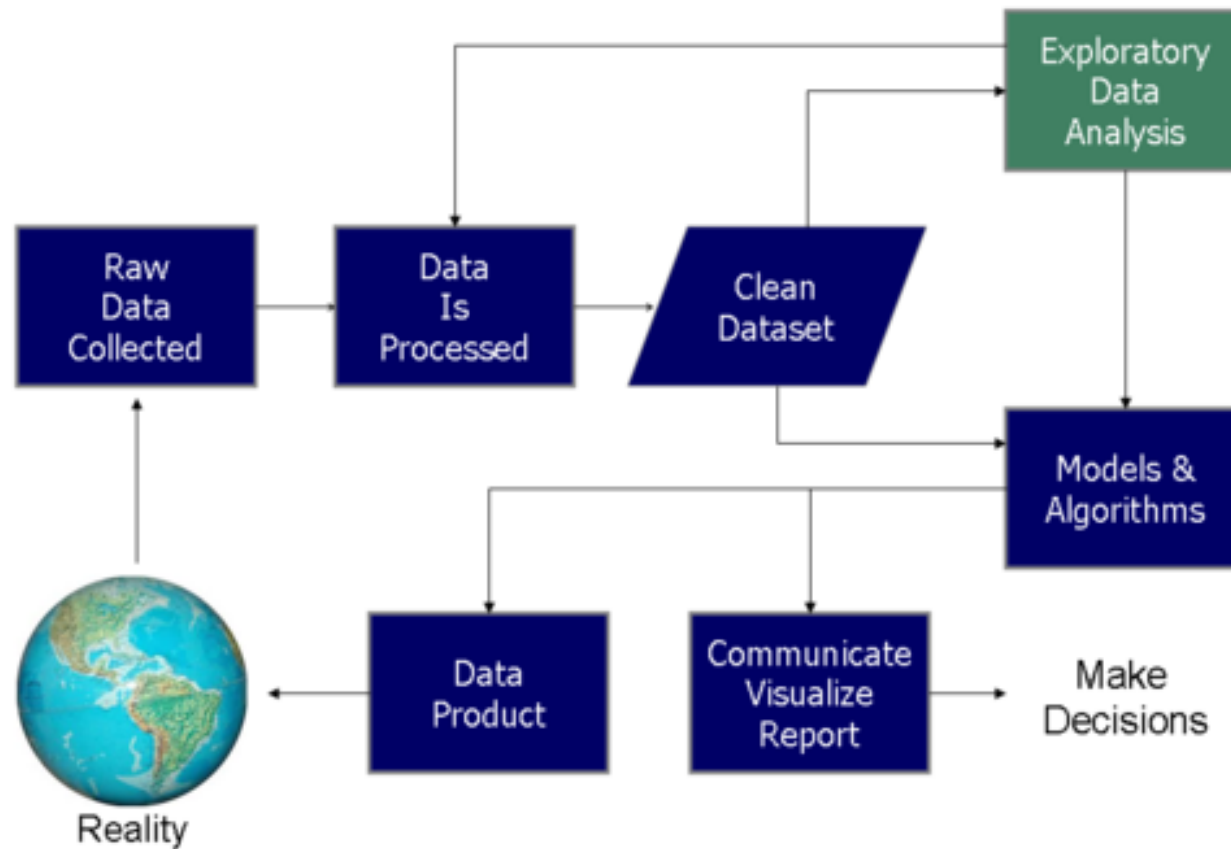
| **String** | Text data, always written inside quotes: `'hello'` or `"Seattle"` |

| **Boolean** | A True/False value, used to evaluate conditions |

| **#** (hash) | Marks a **comment** — Python ignores everything after it; comments are notes for the reader |

THE DATA SCIENCE PROCESS

Overview of the key steps in the data science workflow.



Source: Flow of data science process, Wikipedia

EXPLORATORY DATA ANALYSIS (EDA)

- EDA involves analyzing data sets to understand data structure, summarize their main characteristics, often using visual methods. It helps to discover patterns, spot anomalies, and test hypotheses.
- Key tasks include:
 - Using statistics to explore data
 - Visualizing data using graphs and charts
 - Identifying trends, correlations, and distributions

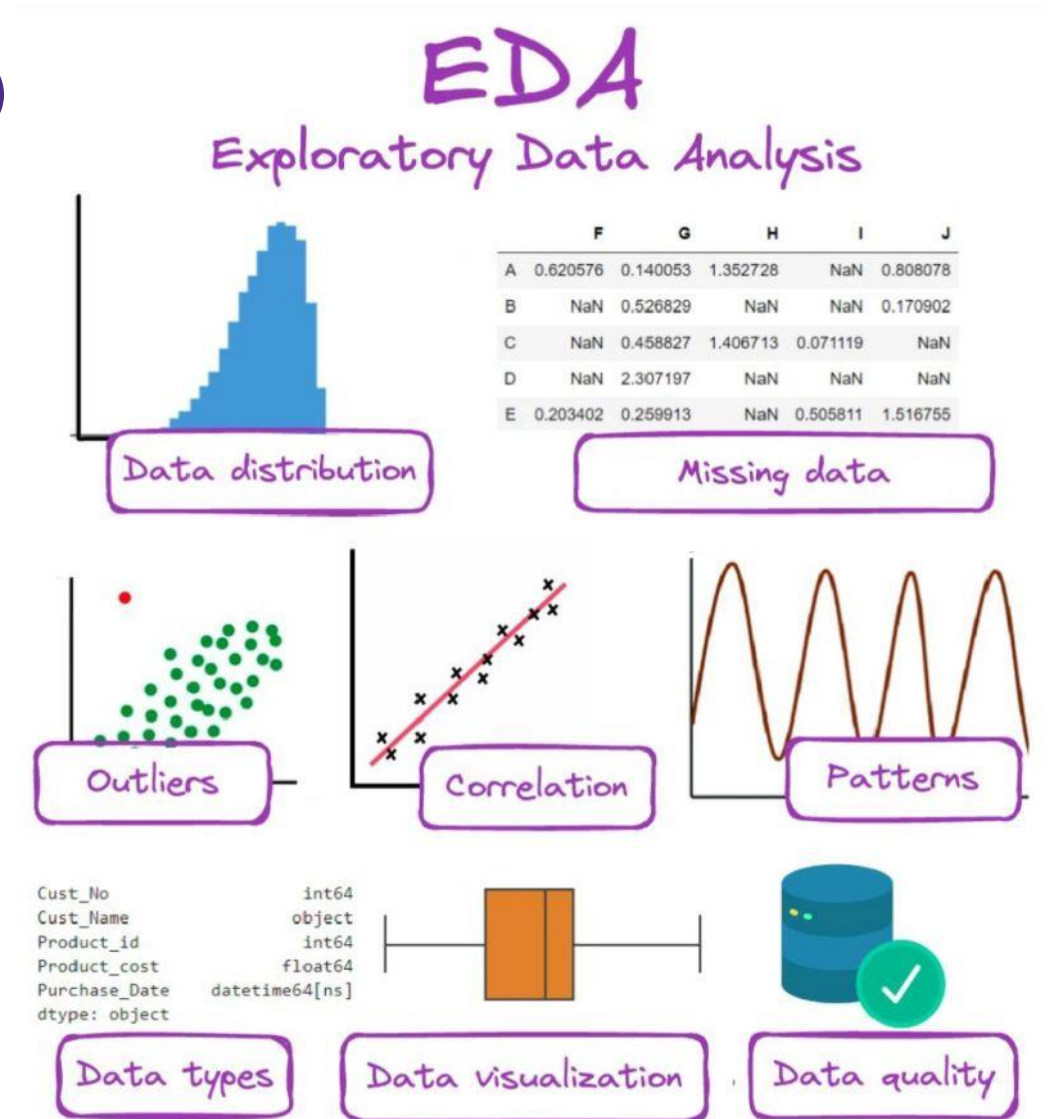


Image Source: <https://www.markovml.com/blog/exploratory-data-analysis>

EXPLORING DATAFRAMES IN PANDAS

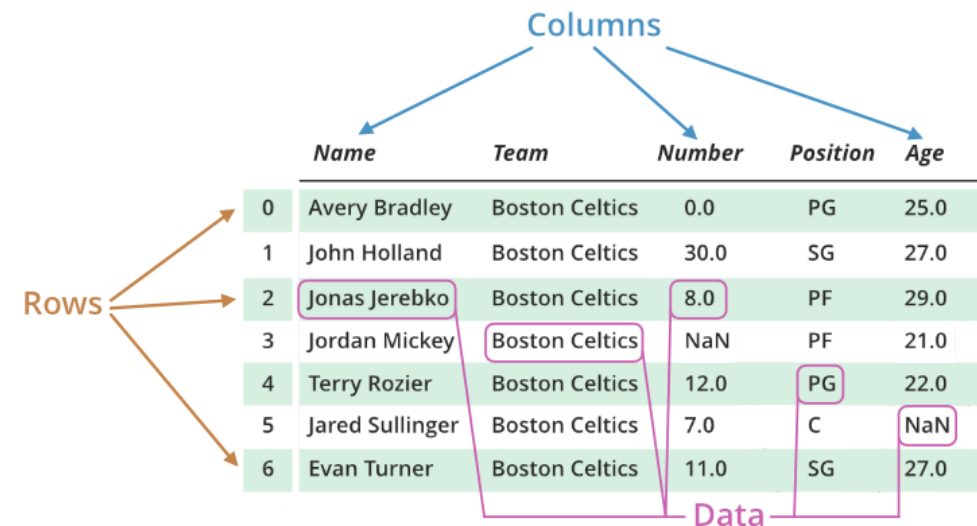
- pandas is a powerful library for data manipulation and analysis in Python. Exploring DataFrames is crucial to understanding the dataset.

- Key functions include:
 - `head()`: Displays the first few rows of the DataFrame.
 - `describe()`: Provides summary statistics for numerical columns.
 - `info()`: Gives a concise summary of the DataFrame.
 - `value_counts()`: Shows the frequency of unique values in a column.

```
import pandas as pd

# list of strings
lst = ['Geeks', 'For', 'Geeks', 'is',
       'portal', 'for', 'Geeks']

# Calling DataFrame constructor on list
df = pd.DataFrame(lst)
print(df)
```



	Name	Team	Number	Position	Age
0	Avery Bradley	Boston Celtics	0.0	PG	25.0
1	John Holland	Boston Celtics	30.0	SG	27.0
2	Jonas Jerebko	Boston Celtics	8.0	PF	29.0
3	Jordan Mickey	Boston Celtics	NaN	PF	21.0
4	Terry Rozier	Boston Celtics	12.0	PG	22.0
5	Jared Sullinger	Boston Celtics	7.0	C	NaN
6	Evan Turner	Boston Celtics	11.0	SG	27.0

Image Source: <https://www.geeksforgeeks.org/python-pandas-dataframe/>

DATA CLEANING AND PREPROCESSING

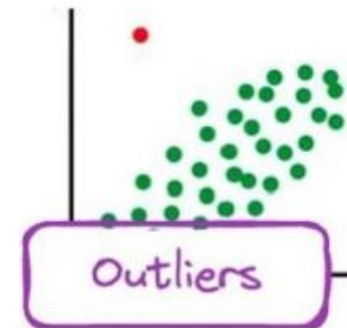
- Data cleaning is the process of identifying and correcting (or removing) inaccurate records from a dataset. Preprocessing includes transforming raw data into an understandable format for analysis.

- Key tasks include:

- Handling missing data
- Removing duplicates
- Removing outliers
- Normalizing and standardizing data
- Encoding categorical variables
- Ensuring data is accurate, consistent, and complete

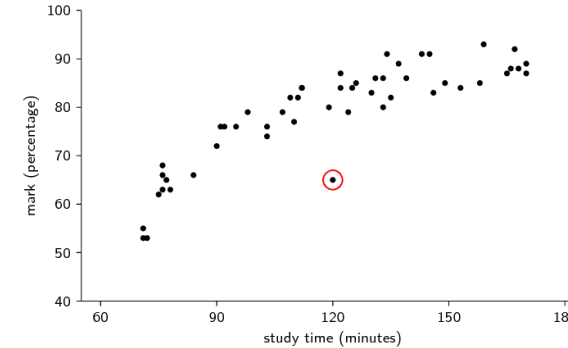
	F	G	H	I	J
A	0.620576	0.140053	1.352728	NaN	0.808078
B	NaN	0.526829	NaN	NaN	0.170902
C	NaN	0.458827	1.406713	0.071119	NaN
D	NaN	2.307197	NaN	NaN	NaN
E	0.203402	0.259913	NaN	0.505811	1.516755

Missing data

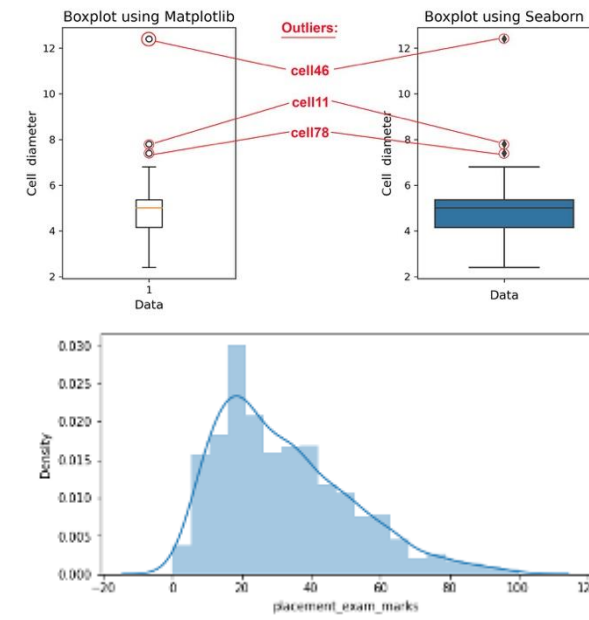


OUTLIERS

- An outlier is a data point that differs significantly from other observations in a dataset. It can occur due to variability in the data or errors in measurement.
- For example, if most buildings consume between 0 and 10 million kWh but one building reports 100 million kWh, that single point can compress the rest of the chart and obscure meaningful patterns. Removing outliers allows us to focus on the majority of the data.
- Key points:
 - Outliers can distort statistical analyses.
 - They may indicate variability or errors.
 - Methods for detecting outliers include z-scores, IQR (Interquartile Range), and visual methods like box plots.



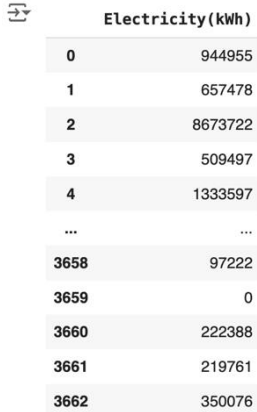
visualizing outliers using boxplots



INDEXING

- Indexing is the process of selecting specific data points or rows from a dataset. It helps in efficiently accessing and manipulating data.
- Key points:
 - Indexing can be done using labels, boolean conditions, or positional indices.
 - Commonly used in dataframes like those in pandas.
 - It allows retrieval of specific rows, columns, or subsets of data.

```
building_energy['Electricity(
```



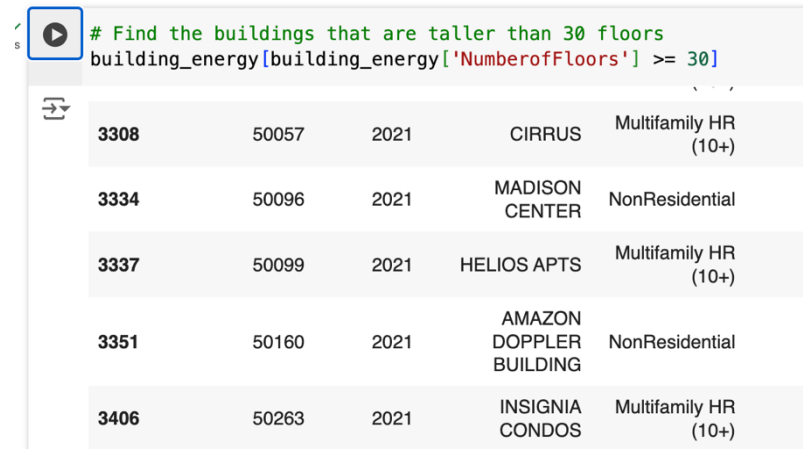
	Electricity(kWh)
0	944955
1	657478
2	8673722
3	509497
4	1333597
...	...
3658	97222
3659	0
3660	222388
3661	219761
3662	350076

3663 rows x 1 columns

dtype: int64

FILTERING

- Filtering is the process of selecting data based on certain conditions or criteria.
- Key points:
 - Filtering is typically used to remove irrelevant or unnecessary data.
 - It can be based on conditions like specific ranges, categories, or values.
 - Commonly used methods include boolean indexing and query-based filtering.



The screenshot shows a Jupyter Notebook interface. At the top, a code cell contains a comment and a pandas query: `# Find the buildings that are taller than 30 floors` and `building_energy[building_energy['NumberOfFloors'] >= 30]`. Below the code cell, a table displays the filtered results. The table has six columns: ID, Energy Star Rating, Year, Building Name, and Type. The filtered results show five buildings: 3308 (CIRRUS), 3334 (MADISON CENTER), 3337 (HELIOS APTS), 3351 (AMAZON DOPPLER BUILDING), and 3406 (INSIGNIA CONDOS). All buildings in the filtered results are from the year 2021.

3308	50057	2021	CIRRUS	Multifamily HR (10+)
3334	50096	2021	MADISON CENTER	NonResidential
3337	50099	2021	HELIOS APTS	Multifamily HR (10+)
3351	50160	2021	AMAZON DOPPLER BUILDING	NonResidential
3406	50263	2021	INSIGNIA CONDOS	Multifamily HR (10+)

MAIN COMMENDS

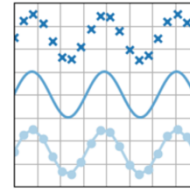
Quick Reference

The following table summarizes the main commands introduced in this notebook.

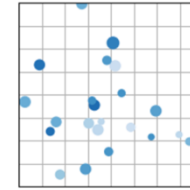
Task	Code
---	---
Load a CSV file	<code>df = pd.read_csv("filename.csv")</code>
Display the first 5 rows	<code>df.head()</code>
List column names	<code>df.columns</code>
Get row and column counts	<code>df.shape</code>
View data types and missing values	<code>df.info()</code>
Compute descriptive statistics	<code>df.describe()</code>
Select a single column	<code>df['ColumnName']</code>
Select rows by position	<code>df.iloc[0:5]</code>
Filter rows by a condition	<code>df[df['Column'] >= value]</code>
Combine filters with AND	<code>df[(condition1) & (condition2)]</code>
Combine filters with OR	<code>df[(condition1) \ (condition2)]</code>
Compute the mean	<code>df['Column'].mean()</code>
Compute min / max	<code>df['Column'].min() / .max()</code>
Create a histogram	<code>plt.hist(df['Column'], bins=30)</code>
Create a scatter plot	<code>plt.scatter(x=df['A'], y=df['B'])</code>
Compute a correlation matrix	<code>df.corr(numeric_only=True)</code>

DATA VISUALIZATION

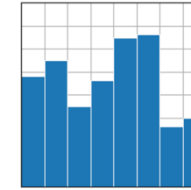
- Data visualization is the representation of data in a visual context, such as a graph or chart, to help understand complex data patterns.
- Common techniques include:
 - Line charts
 - Bar charts
 - Histograms
 - Scatter plots
 - Heatmaps
 - Dashboards



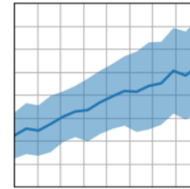
plot(x, y)



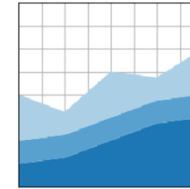
scatter(x, y)



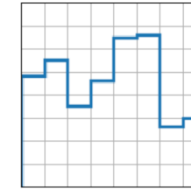
bar(x, height)



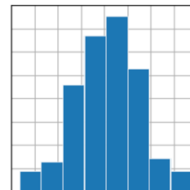
fill_between(x, y1, y2)



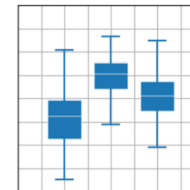
stackplot(x, y)



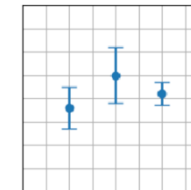
stairs(values)



hist(x)



boxplot(X)



errorbar(x, y, yerr, xerr)

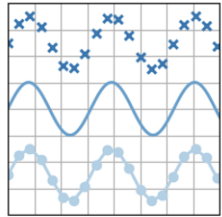
Image Source: matplotlib, https://matplotlib.org/stable/plot_types/index.html

PLOT TYPES

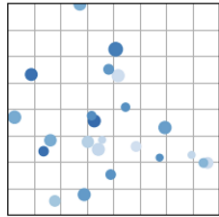
Overview of many common plotting commands provided by matplotlib.

Pairwise data

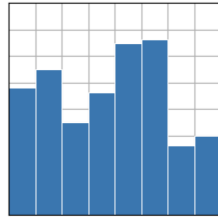
Plots of pairwise (x, y) , tabular $(var_0, \dots, var_n)$, and functional $f(x) = y$ data.



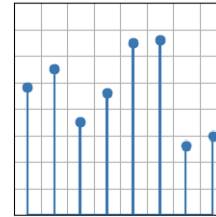
`plot(x, y)`



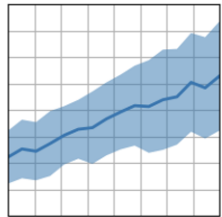
`scatter(x, y)`



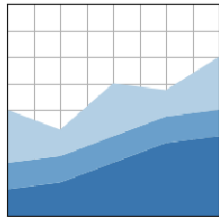
`bar(x, height)`



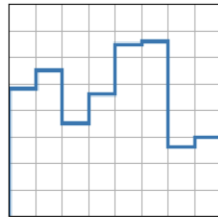
`stem(x, y)`



`fill_between(x, y1,
y2)`



`stackplot(x, y)`



`stairs(values)`

https://matplotlib.org/stable/plot_types/index.html

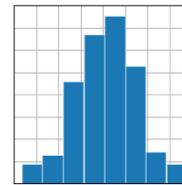
PLOT TYPES

Overview of many common plotting commands provided by matplotlib.

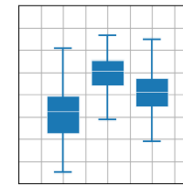


Statistical distributions

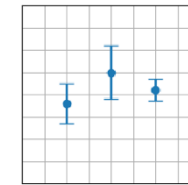
Plots of the distribution of at least one variable in a dataset. Some of these methods also compute the distributions.



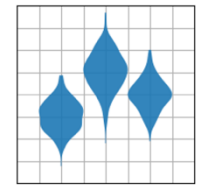
hist(x)



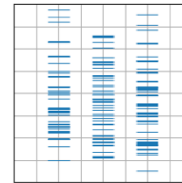
boxplot(X)



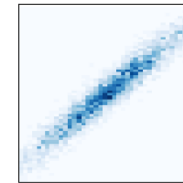
errorbar(x, y, yerr,
xerr)



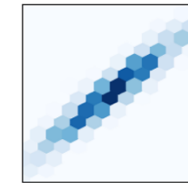
violinplot(D)



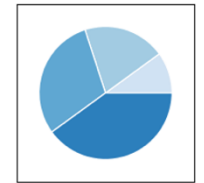
eventplot(D)



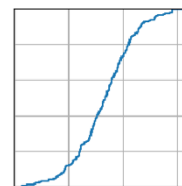
hist2d(x, y)



hexbin(x, y, C)



pie(x)



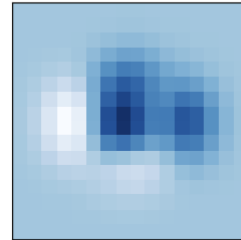
ecdf(x)

PLOT TYPES

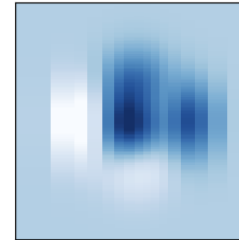
Overview of many common plotting commands provided by matplotlib.

Gridded data

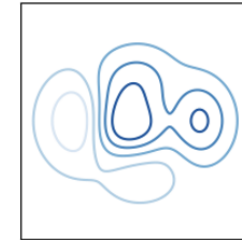
Plots of arrays and images $Z_{i,j}$ and fields $U_{i,j}$, $V_{i,j}$ on regular grids and corresponding coordinate grids $X_{i,j}$, $Y_{i,j}$.



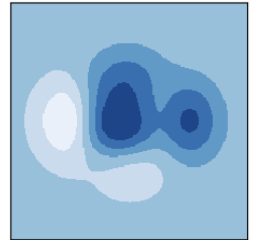
imshow(Z)



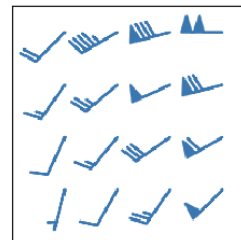
pcolormesh(X, Y, Z)



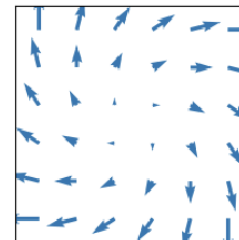
contour(X, Y, Z)



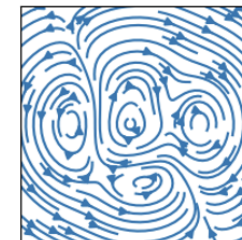
contourf(X, Y, Z)



barbs(X, Y, U, V)



quiver(X, Y, U, V)



streamplot(X, Y, U, V)



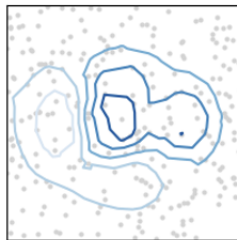
https://matplotlib.org/stable/plot_types/index.html

PLOT TYPES

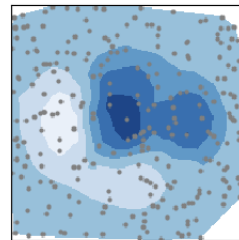
Overview of many common plotting commands provided by matplotlib.

Irregularly gridded data

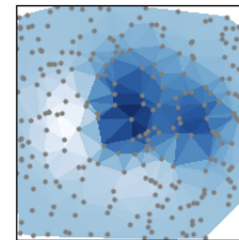
Plots of data $Z_{x,y}$ on unstructured grids, unstructured coordinate grids (x, y) , and 2D functions $f(x, y) = z$.



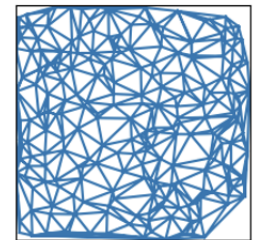
`tricontour(x, y, z)`



`tricontourf(x, y, z)`



`tripcolor(x, y, z)`



`triplot(x, y)`

matplotlib

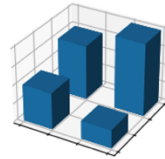
https://matplotlib.org/stable/plot_types/index.html

PLOT TYPES

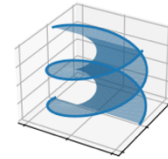
Overview of many common plotting commands provided by matplotlib.

3D and volumetric data

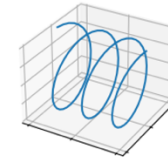
Plots of three-dimensional (x, y, z) , surface $f(x, y) = z$, and volumetric $V_{x,y,z}$ data using the `mpl_toolkits.mplot3d` library.



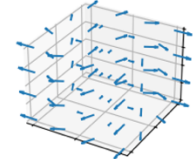
`bar3d(x, y, z, dx, dy, dz)`



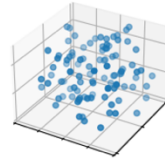
`fill_between(x1, y1, z1, x2, y2, z2)`



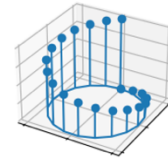
`plot(xs, ys, zs)`



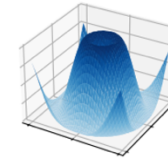
`quiver(X, Y, Z, U, V, W)`



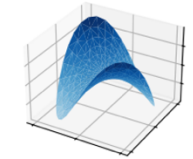
`scatter(xs, ys, zs)`



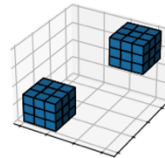
`stem(x, y, z)`



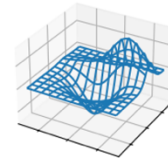
`plot_surface(X, Y, Z)`



`plot_trisurf(x, y, z)`



`voxels([x, y, z], filled)`



`plot_wireframe(X, Y, Z)`

matplotlib

https://matplotlib.org/stable/plot_types/index.html

STATISTICS

- Statistics is the study of data collection, analysis, interpretation, and presentation.
 - Two types: **Descriptive** and **Inferential**.
 - Descriptive: Summarizing data.
 - Inferential: Drawing conclusions from data.

DESCRIPTIVE STATISTICS

- **Measures of Central Tendency:** Mean, Median, Mode.
- **Measures of Spread:** Variance, Standard Deviation, Range.
- **Visualization Tools:** Histograms, Boxplots, etc.

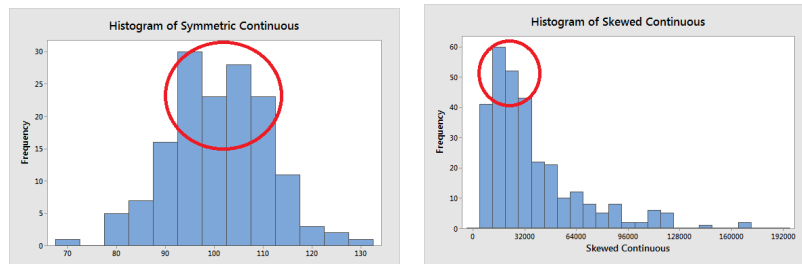


Image Source: <https://statisticsbyjim.com/basics/measures-central-tendency-mean-median-mode/>

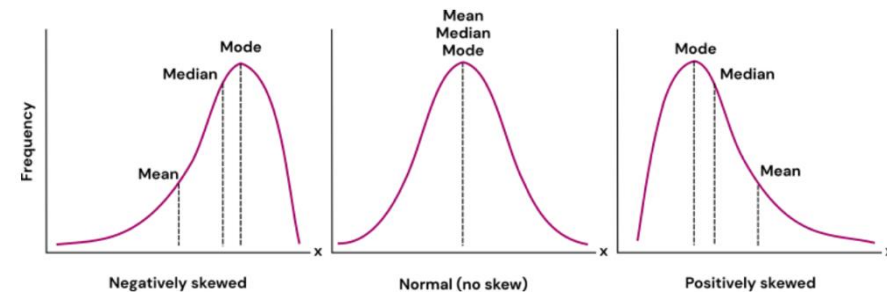


Image Source: <https://ledidi.com/academy/measures-of-central-tendency-mean-median-and-mode>

DESCRIPTIVE STATISTICS

- **Measures of Central Tendency:**

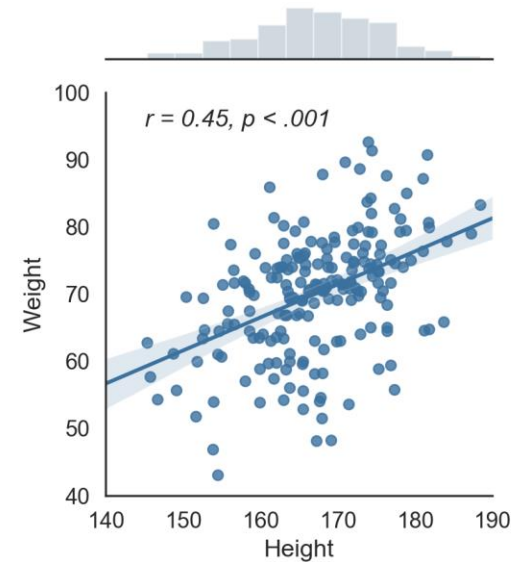
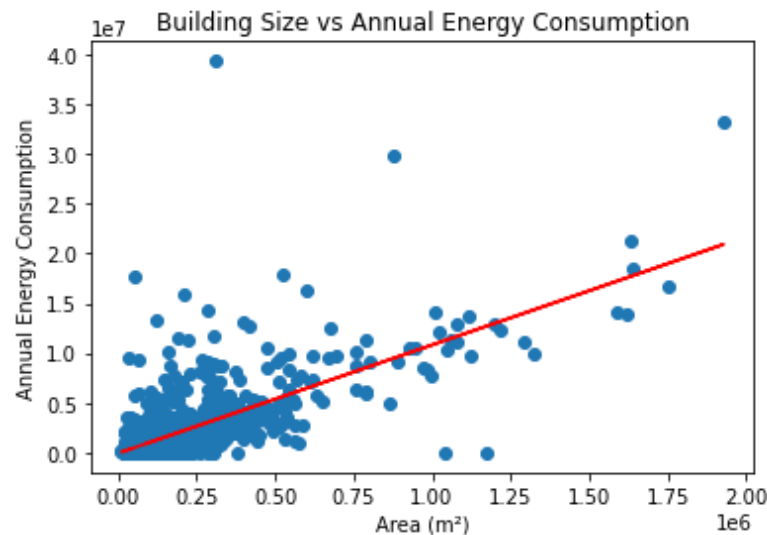
- **Mean:** The average of a set of numbers. Add them up and divide by how many numbers there are.
- **Median:** The middle value when you arrange the numbers in order. If there's an even number of values, it's the average of the two middle ones.
- **Mode:** The number that appears most often in a set of numbers.

- **Measures of Spread:**

- **Variance:** Tells us how spread out the numbers are. It's the average of the squared differences between each number and the mean.
- **Standard Deviation:** A measure of how spread out the numbers are, similar to variance but easier to interpret because it's in the same units as the data.
- **Range:** The difference between the largest and smallest numbers in a set. It shows how wide the data is spread.

CORRELATION

- Correlation is a statistical technique that measures the strength of the relationship between two variables.
- It is used to understand how one variable may predict or be associated with another.



CORRELATION

Direction of Relationship:

- **Positive Correlation:** When one variable increases, the other variable tends to increase as well. For example, height and weight are typically positively correlated in adults.
- **Negative Correlation:** When one variable increases, the other variable tends to decrease. For example, the amount of time spent studying and the number of errors on a test might be negatively correlated.

Strength of Relationship:

- The correlation coefficient (usually denoted as "r") measures the strength of the relationship between the variables. It ranges from -1 to +1, where:
 - +1 indicates a perfect positive correlation.
 - 0 indicates no correlation.
 - -1 indicates a perfect negative correlation.

▶ # Correlation matrix for numeric columns

```
df_noout.corr(numeric_only=True)
```

...	OSEBuildingID	DataYear	ZipCode	Latitude	Longitude	CouncilDistrictCode	YearBuilt	NumberofFloors	NumberofBuildings	PropertyGFATotal	...
OSEBuildingID	1.000000	NaN	-0.224928	-0.021115	0.025804	0.001514	0.347554	0.018901	-0.016079	-0.070071	...
DataYear	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...
ZipCode	-0.224928	NaN	1.000000	-0.003425	0.003019	0.063447	-0.135127	-0.038241	0.011985	-0.021946	...
Latitude	-0.021115	NaN	-0.003425	1.000000	-0.999070	0.040109	-0.014764	0.009265	0.003248	0.000651	...
Longitude	0.025804	NaN	0.003019	-0.999070	1.000000	-0.029600	0.019340	-0.010737	-0.002402	-0.002510	...
CouncilDistrictCode	0.001514	NaN	0.063447	0.040109	-0.029600	1.000000	-0.088721	-0.034142	-0.038188	-0.133396	...
YearBuilt	0.347554	NaN	-0.135127	-0.014764	0.019340	-0.088721	1.000000	0.184357	0.005401	0.203338	...
NumberofFloors	0.018901	NaN	-0.038241	0.009265	-0.010737	-0.034142	0.184357	1.000000	-0.027242	0.629509	...
NumberofBuildings	-0.016079	NaN	0.011985	0.003248	-0.002402	-0.038188	0.005401	-0.027242	1.000000	0.262911	...
PropertyGFATotal	-0.070071	NaN	-0.021946	0.000651	-0.002510	-0.133396	0.203338	0.629509	0.262911	1.000000	...
PropertyGFABuilding(s)	-0.085086	NaN	-0.021691	-0.001897	-0.000354	-0.130610	0.170213	0.626217	0.282521	0.977635	...
PropertyGFAParking	-0.007364	NaN	-0.016697	0.007872	-0.008081	-0.105079	0.244005	0.467272	0.134238	0.792023	...
ENERGYSTARScore	0.112552	NaN	-0.087157	0.034352	-0.031692	0.043843	0.196330	0.169368	-0.056066	0.099563	...
SiteEUIWN(kBtu/sf)	-0.095698	NaN	0.016663	0.004290	-0.004192	-0.027014	-0.066702	-0.039622	0.061263	0.044423	...
SiteEUI(kBtu/sf)	-0.095075	NaN	0.017873	0.004101	-0.004015	-0.029867	-0.065098	-0.040733	0.061003	0.045415	...
SiteEnergyUse(kBtu)	-0.123297	NaN	0.005443	0.000708	-0.001650	-0.107875	0.088051	0.351826	0.305858	0.684523	...
SiteEnergyUseWN(kBtu)	-0.121640	NaN	0.004949	0.000763	-0.001726	-0.106636	0.088310	0.351061	0.306118	0.682904	...
SourceEUIWN(kBtu/sf)	-0.111244	NaN	0.015327	0.005297	-0.004665	-0.036604	0.007721	-0.012116	0.064835	0.066221	...
SourceEUI(kBtu/sf)	-0.110077	NaN	0.016415	0.005104	-0.004471	-0.038604	0.009944	-0.012314	0.064631	0.067538	...
LargestPropertyUseTypeGFA	-0.084715	NaN	-0.018455	-0.003084	0.001303	-0.125239	0.181474	0.635012	0.260790	0.963370	...
ondLargestPropertyUseTypeGFA	-0.027406	NaN	-0.024264	0.003947	-0.005198	-0.114152	0.211941	0.484008	0.165269	0.823309	...
irdLargestPropertyUseTypeGFA	-0.005830	NaN	0.005375	0.003433	-0.005078	-0.085630	0.092294	0.254966	0.076083	0.416091	...
Electricity(kWh)	-0.141803	NaN	0.000512	0.001070	-0.002357	-0.122676	0.133807	0.455171	0.264415	0.751536	...

CORRELATION

NOTE:

Causation is Not Correlation: A crucial aspect of correlation is that it does not imply causation.

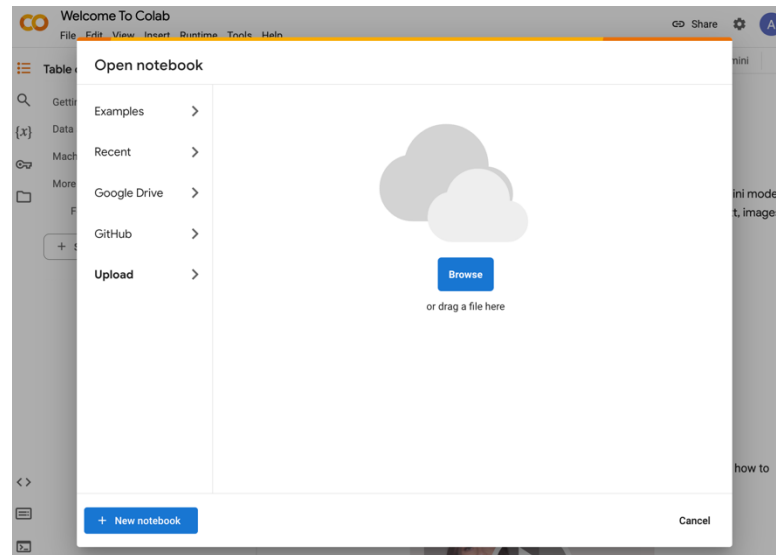
Two variables may be correlated without one causing the other. They could be influenced by a third factor, or the correlation might be coincidental.

SUMMARY

- Exploratory Data Analysis (EDA): Understanding patterns and anomalies in data.
- Pandas DataFrames: Practical and efficient data manipulation and analysis.
- Data Collection, Cleaning, and Preparation: Ensuring data accuracy and readiness.
- Data Visualization: Uncovering trends and communicating insights.
- Key Statistical Concepts: Interpreting data with foundational statistics.
- Correlation: Identifying relationships between variables.

LAB ACTIVITY: EXPLORATORY DATA ANALYSIS

1. Open up a web browser. Google “google colab” click on the first entry. You may also access the website by following <https://colab.research.google.com/>
2. Go to Canvas and download the file called “Exploratory_Data_Analysis_Pandas.ipynb”



NEXT CLASS

- Exploratory Data Analysis

